Evaluation of First Aid Awareness Among a Sample of the Alexandrian Community

(An Analytical Field Study Merging Historical Roots with Digital Logic – CS50 Inspired)

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Academic Year: 2025 – 2026

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1. Abstract

This study aims to measure the level of first aid awareness among a sample of (41) individuals in Alexandria. An electronic survey was utilized to evaluate knowledge regarding emergency situations, including fainting, burns, bleeding, and the national emergency number (123). Results indicated a high level of awareness in managing bleeding at 79.5%, while awareness significantly decreased regarding fainting protocols (56.4%) and knowledge of the emergency hotline (57.5%). The study concludes with the necessity of integrating technical solutions to enhance the efficiency of medical emergency response.

2. Introduction

First aid serves as the vital primary "Algorithm" for saving lives. In the world of programming, a syntax error might crash an application; however, in medicine, a procedural error in first aid can lead to critical health deterioration. First aid represents the frontline defense in emergencies, where correct intervention is a decisive factor in limiting complications before specialized care arrives.

3. Historical Overview: From Traditional Practice to the Digital Age

- **Ancient Roots: Ancient Egyptians and Greeks utilized bandages and linen to control bleeding. Organized practice emerged with the "Knights of St. John" in the 11th century.**
- **Historical Milestones: 1859 marked the founding of the International Committee of the Red Cross by Henry Dunant. In 1877, Friedrich von Esmarch coined the modern term "First Aid".**
- **Technical Vision (CS50 Perspective): We are currently transitioning from the era of manual skill alone to the era of "Smart First Aid". Programming languages and algorithms now contribute to reducing response times and guiding first responders accurately through Artificial Intelligence (AI) and Big Data applications.**

4. Methodology

The study adopted a descriptive-analytical approach supported by logical thinking inspired by the CS50 curriculum:

- **Research Tool: An interactive digital survey distributed via social media platforms.**
- **Sample: (41) participants from the Alexandria Governorate.**

- **Data Analysis: Responses were treated as a Data Array to extract precise percentages.**

5. Results & Statistical Analysis

I.

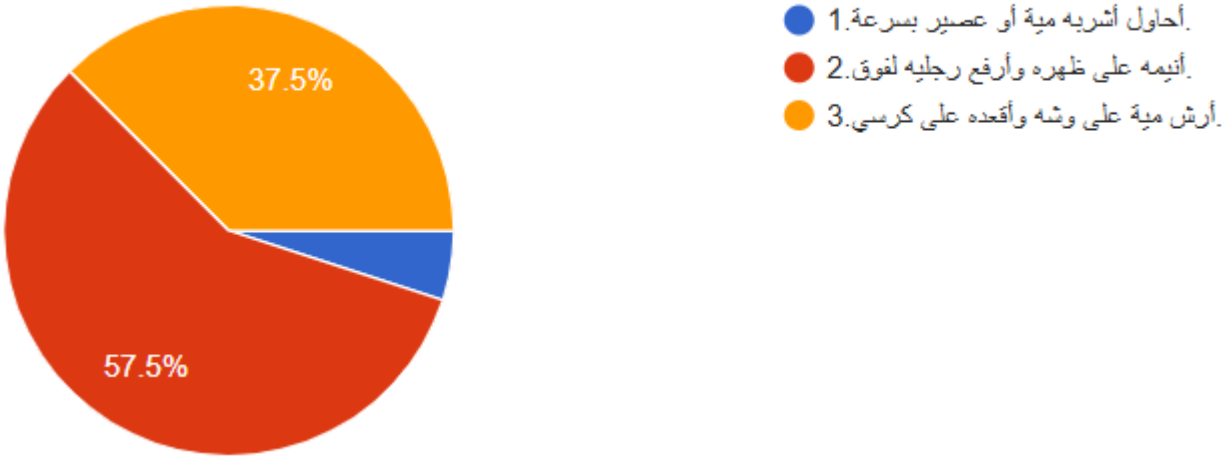

Figure 1: Community response to fainting scenarios. The chart shows that **57.5%** of participants correctly identified the recovery position, while **37.5%** still rely on incorrect traditional methods like splashing water on the face

Management of Fainting Cases

- **Analysis: Results show that 57.5% of the sample chose the correct procedure, while 37.5% leaned toward incorrect practices, such as spraying water on the face.**
- **Programming Perspective: This reflects a bug in the "Behavioral Algorithm" that requires cognitive debugging.**

II.

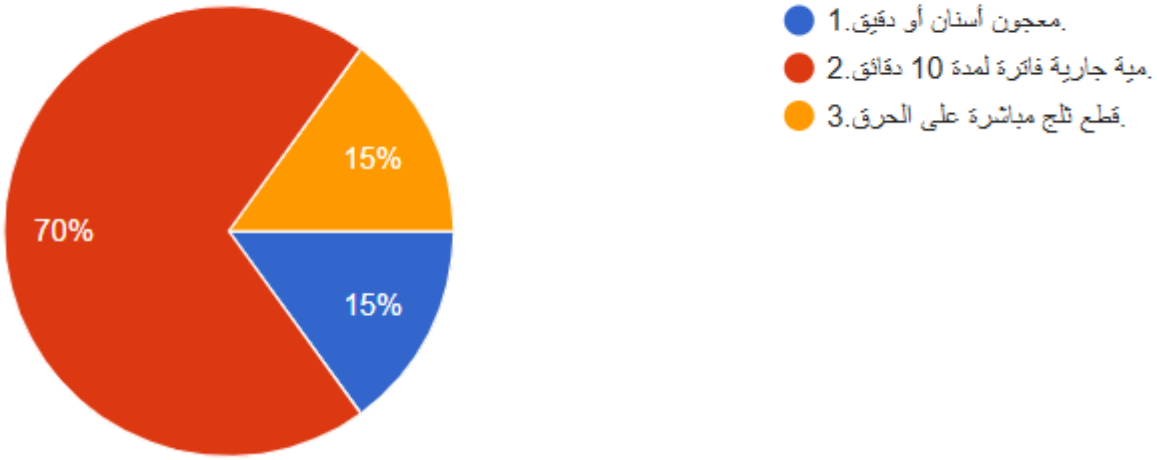

Figure 2: Knowledge of minor burn management. Results indicate a positive awareness level, with **70%** of the sample choosing running water as the primary treatment, reflecting a decline in common misconceptions.

Management of Minor Burns

- **Analysis: Results showed an accuracy rate of 70% in favor of using running water, which is a positive indicator of the decline of certain misconceptions within the Alexandrian community.**

III.

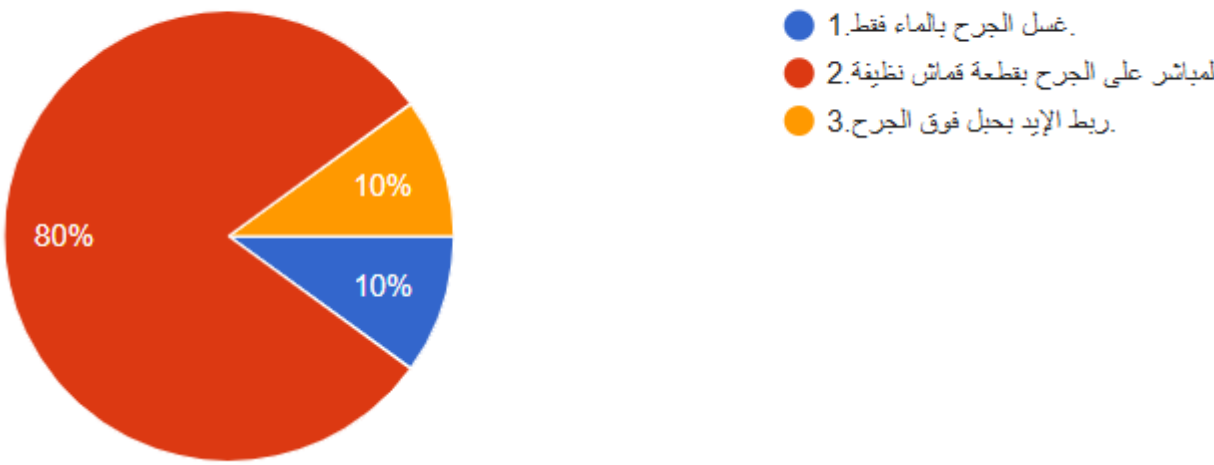

Figure 3: Efficiency in managing open wounds and bleeding. This category achieved the highest accuracy at **80%**, demonstrating a solid understanding of how to apply direct pressure to stop bleeding.

Management of Bleeding Cases

- **Analysis: The highest success rate was recorded in this category at 80%, proving a stable knowledge database among the sample regarding open wound management.**

IV.

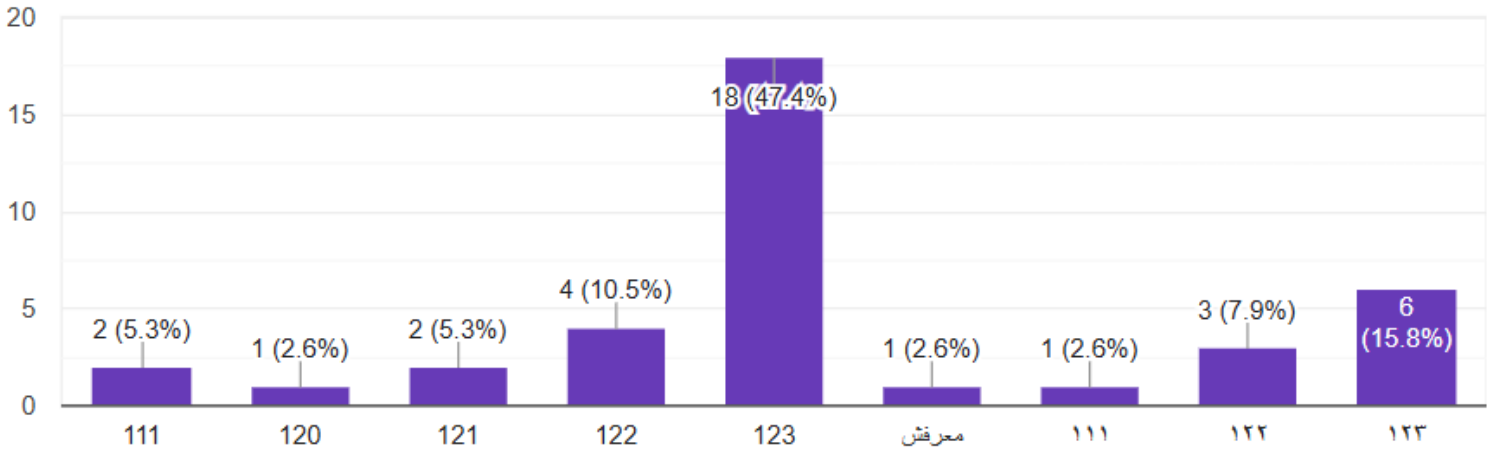

Figure 4: Awareness of the National Emergency Number (123). The data reveals a critical gap, as only **47.4%** correctly identified the number, which represents a "Broken Link" in the emergency response chain.

Awareness of the National Emergency Hotline

- **Analysis: The data reveals significant dispersion; only 47.4% correctly identified the national number.**
- **Technical Interpretation: This represents a "Broken Link" in the response system. Regardless of the responder's skill level, a failure to connect with the system halts the entire rescue process.**
- **Future Directions & Technical Integration** To bridge the gap between medical awareness and practical application, this research was integrated with a computational logic model. A Python-based decision tree was developed to simulate emergency responses based on the findings of this study. The full source code, data analysis, and project documentation are available on GitHub for further academic review and development.
- **Project Repository:** https://github.com/mina-s-zakaria2010-sketch/Medical-Logic-CS50

6. Conclusion & Recommendations

The study demonstrates that awareness requires systematic reinforcement merging medicine and technology. The researcher recommends:

- **Technically: Developing applications based on "Decision Trees" to guide users step-by-step during emergencies.**
- **Socially: Intensifying first aid training courses and establishing "123" as the sole protocol for emergency rescue.**

7. References

1. **World Health Organization (2018). First Aid Manual.**
2. **International Federation of Red Cross and Red Crescent Societies (2020).**
3. **Egyptian Ministry of Health and Population – Emergency Services (123).**